

Today's tutorial is attendance-based. Please

- 1) sign the attendance sheet with TAs
- 2) submit your codes.

Task:

The code of merge sort is given. It generates a controlled set of N random numbers and then apply merge sort to it. The output is a sorted array. Now, modify this code so that

- 1) It outputs the indices (0 to N-1) of the array instead of the value.
- 2) The indices should be sorted such that the corresponding values in the array are in descending order.
- 3) You must not modify the original array (assume it is read-only).

Example:

Input: arr = [50, 20, 30, 10]

Output: [3, 1, 2, 0]

Explanation:

arr[0] = 50

arr[1] = 20

arr[2] = 30

arr[3] = 10

The sorted array is [10, 20, 30, 50], so the corresponding indices are [3, 1, 2, 0]